Web Security, SSL and TLS

Overview

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- SSL Architecture
- Cryptographic Computations
- Introduction to Transport Layer Security (TLS)
- HTTPS
- Secure Shell (SSH)
- Summary

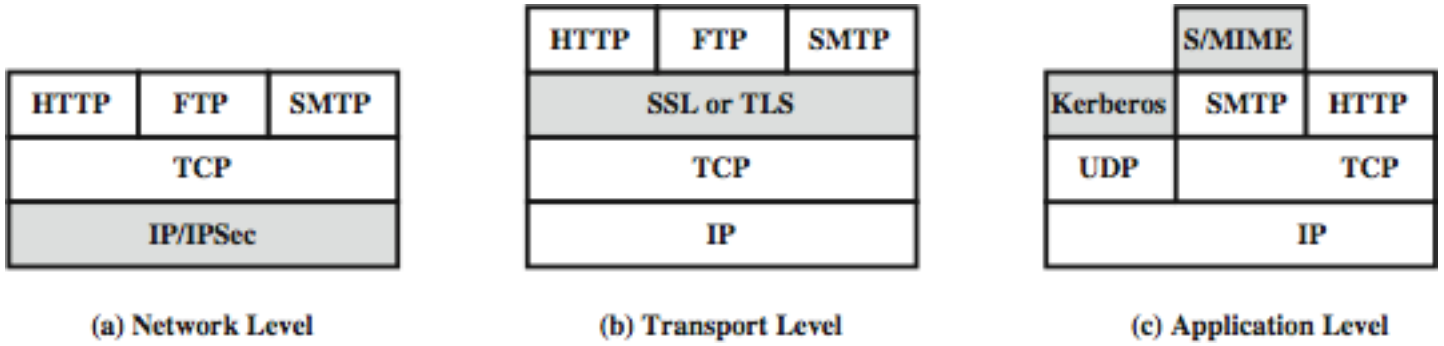
Introduction to Web Security

- Web now widely used by business, government, individuals
- but Internet & Web are vulnerable
- have a variety of threats
  - integrity
  - confidentiality
  - denial of service
  - authentication
- need added security mechanisms

A Comparison of Threats on the Web

	Threats	Consequences	Countermeasures
Integrity	• Modification of user data • Trojan horse browser • Modification of memory • Modification of message traffic in transit	• Loss of information • Compromise of machine • Vulnerability to all other threats	Cryptographic checksums
Confidentiality	• Eavesdropping on the net • Theft of info from server • Theft of data from client • Info about network configuration • Info about which client talks to server	• Loss of information • Loss of privacy	Encryption, Web proxies
Denial of Service	• Killing of user threads • Flooding machine with bogus requests • Filling up disk or memory • Isolating machine by DNS attacks	• Disruptive • Annoying • Prevent user from getting work done	Difficult to prevent
Authentication	• Impersonation of legitimate users • Data forgery	• Misrepresentation of user • Belief that false information is valid	Cryptographic techniques

Web Traffic Security Approaches



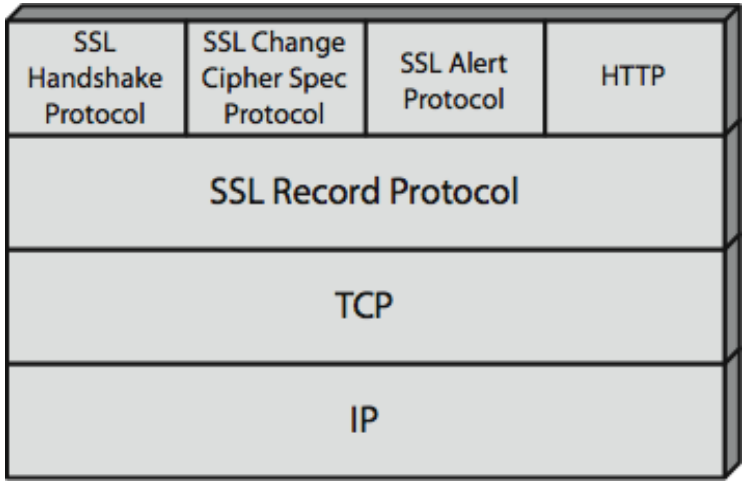
- The advantage of using IPsec is that it is transparent to end users and applications and provides a general-purpose solution.
- IPsec includes a filtering capability so that only selected traffic need incur the overhead of IPsec processing.
- For full generality, SSL (or TLS) could be provided as part of the underlying protocol suite and therefore be transparent to applications.
- TLS can be embedded in specific packages.
  - ✓ Virtually all browsers come equipped with TLS, and most Web servers have implemented the protocol.
- Application-specific security services are embedded within the particular application.
- The service can be tailored to the specific needs of a given application.

Secure Socket Layer (SSL)

- transport layer security service
- originally developed by Netscape
- version 3 designed with public input
- subsequently became Internet standard known as TLS (Transport Layer Security)
- uses TCP to provide a reliable end-to-end service
- SSL has two layers of protocols

SSL Architecture

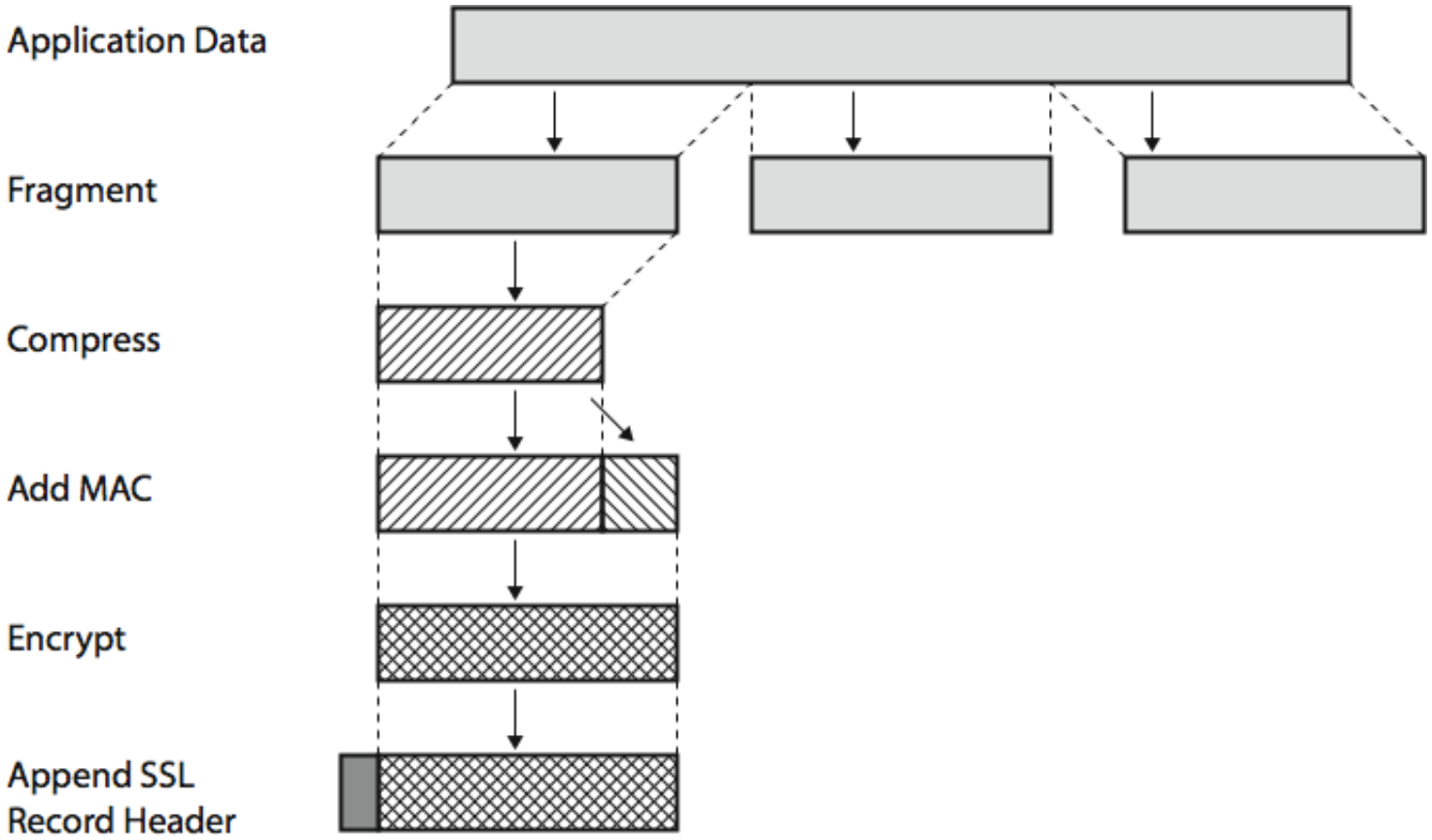
- **SSL connection**
  - a transient, peer-to-peer, communications link
  - associated with SSL session
- **SSL session**
  - an association between client & server
  - created by the Handshake Protocol
  - define a set of cryptographic parameters
  - may be shared by multiple SSL connections



SSL Record Protocol Services

- **Confidentiality**
  - using symmetric encryption with a shared secret key defined by Handshake Protocol
  - AES, IDEA, RC2-40, DES-40, DES, 3DES, Fortezza, RC4-40, RC4-128
  - message is compressed before encryption
- **Message integrity**
  - using a MAC with shared secret key
  - similar to HMAC but with different padding

SSL Record Protocol Operation



SSL Change Cipher Spec Protocol

- one of 3 SSL specific protocols which use the SSL Record protocol
- a single message
- causes pending state to become current
- hence updating the cipher suite in use

(a) Change Cipher Spec Protocol

SSL Alert Protocol

- conveys SSL-related alerts to peer entity
- severity
  - warning or fatal
- specific alert
  - fatal: unexpected message, bad record mac, decompression failure, handshake failure, illegal parameter
  - warning: close notify, no certificate, bad certificate, unsupported certificate, certificate revoked, certificate expired, certificate unknown
- compressed & encrypted like all SSL data

1 byte	1 byte
Level	Alert

(b) Alert Protocol

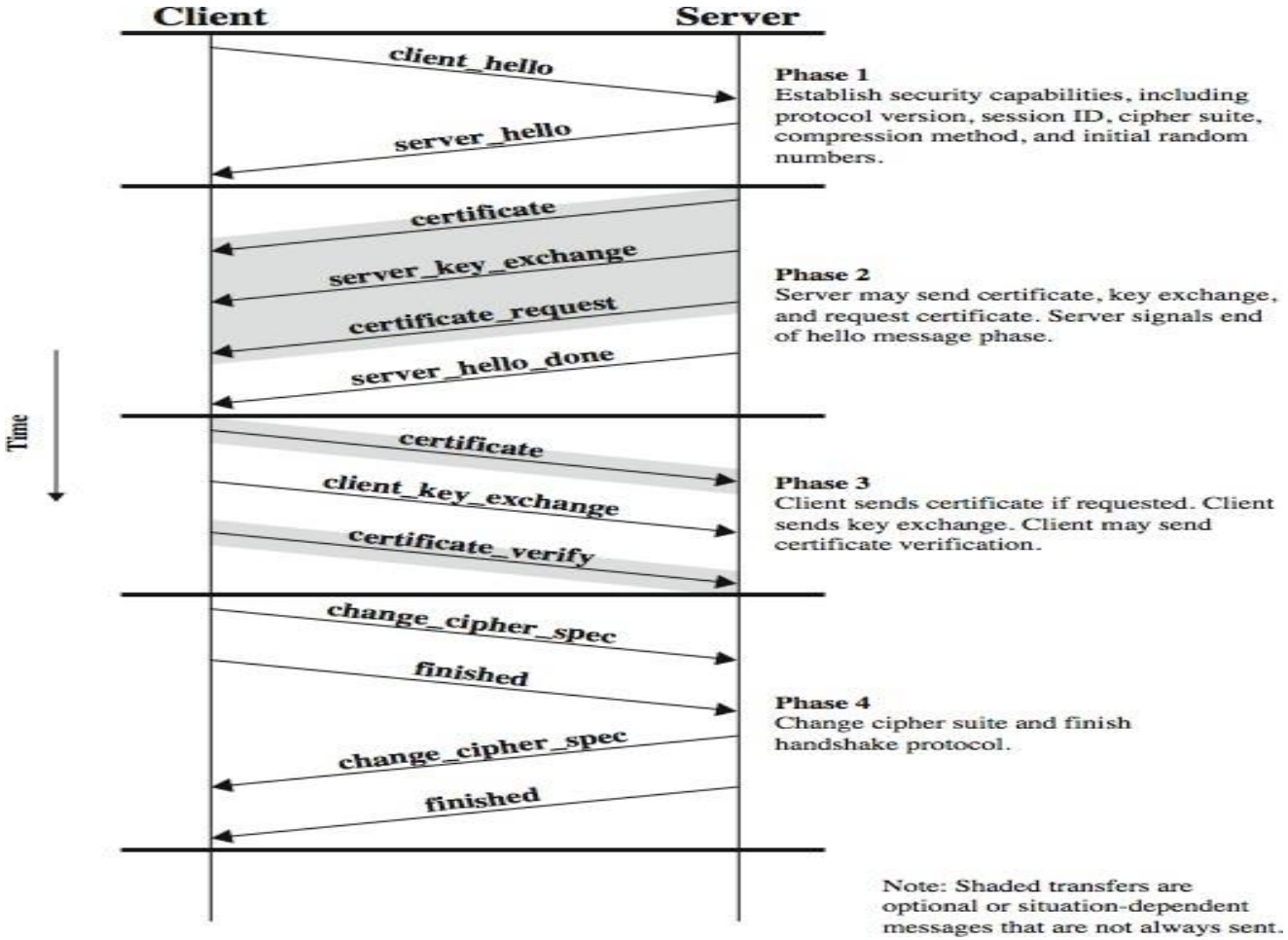
SSL Handshake Protocol

- allows server & client to:
  - authenticate each other
  - to negotiate encryption & MAC algorithms
  - to negotiate cryptographic keys to be used
- comprises a series of messages in phases
  - Establish Security Capabilities
  - Server Authentication and Key Exchange
  - Client Authentication and Key Exchange
  - Finish

1 byte	3 bytes	≥ 0 bytes
Type	Length	Content

(c) Handshake Protocol

SSL Handshake Protocol … Contd.



Cryptographic Computations

- master secret creation
  - a one-time 48-byte value
  - generated using secure key exchange (RSA / Diffie-Hellman) and then hashing info
- generation of cryptographic parameters
  - client write MAC secret, a server write MAC secret, a client write key, a server write key, a client write IV, and a server write IV
  - generated by hashing master secret

Introduction to Transport Layer Security (TLS)

- IETF standard RFC 2246 similar to SSLv3
- with minor differences
  - in record format version number
  - uses HMAC for MAC
  - a pseudo-random function expands secrets
    - based on HMAC using SHA-1 or MD5
  - has additional alert codes
  - some changes in supported ciphers
  - changes in certificate types & negotiations
  - changes in crypto computations & padding

HTTPS

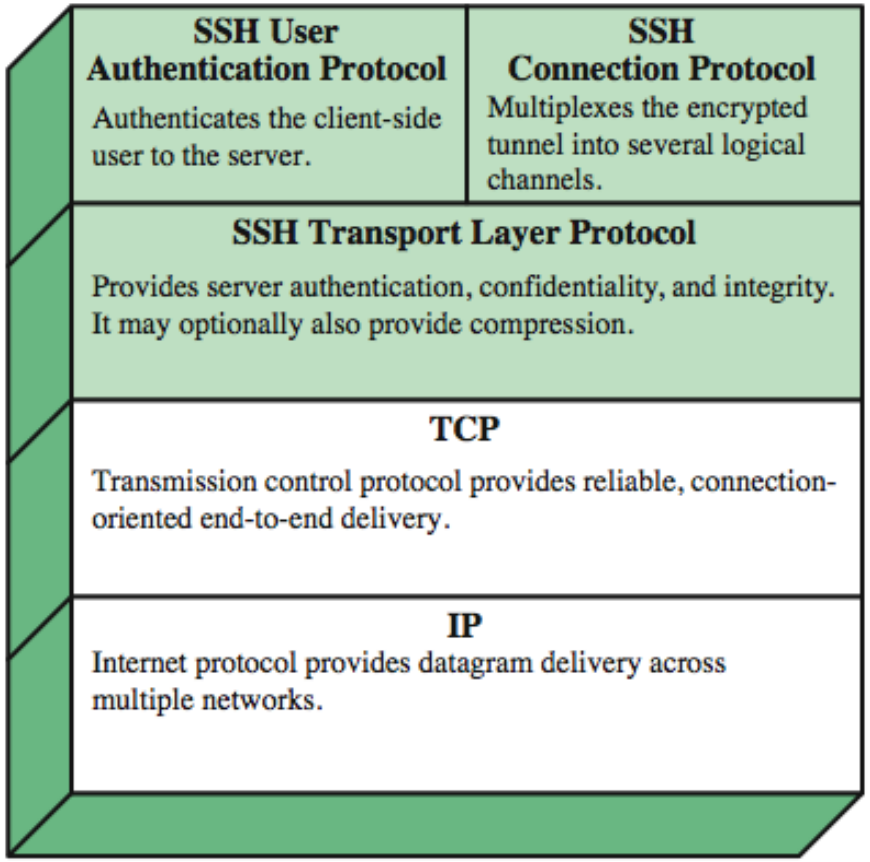
- HTTPS (HTTP over SSL)
  - combination of HTTP & SSL/TLS to secure communications between browser & server
    - documented in RFC2818
    - no fundamental change using either SSL or TLS
- use https:// URL rather than http://
  - and port 443 rather than 80
- encrypts
  - URL, document contents, form data, cookies, HTTP headers

HTTPS Use

- connection initiation
  - TLS handshake then HTTP request(s)
- connection closure
  - have “Connection: close” in HTTP record
  - TLS level exchange close_notify alerts
  - can then close TCP connection
  - must handle TCP close before alert exchange sent or completed

Secure Shell (SSH)

- protocol for secure network communications
  - designed to be simple & inexpensive
- SSH1 provided secure remote logon facility
  - replace TELNET & other insecure schemes
  - also has more general client/server capability
- SSH2 fixes a number of security flaws
- documented in RFCs 4250 through 4254
- SSH clients & servers are widely available
- method of choice for remote login/ X tunnels



SSH Transport Layer Protocol

- server authentication occurs at transport layer, based on server/host key pair(s)
  - server authentication requires clients to know host keys in advance
- packet exchange
  - establish TCP connection
  - can then exchange data
    - identification string exchange, algorithm negotiation, key exchange, end of key exchange, service request
  - using specified packet format

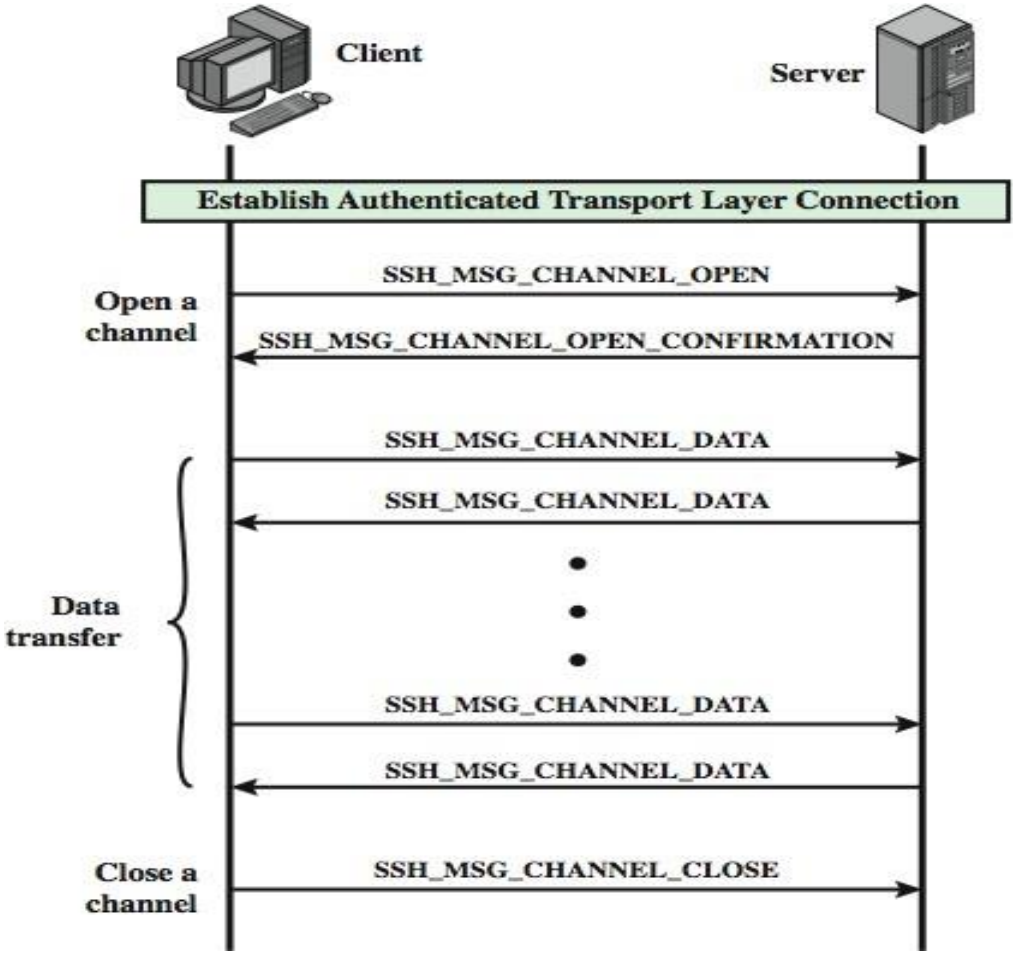
SSH User Authentication Protocol

- authenticates client to server
- three message types:
  - SSH_MSG_USERAUTH_REQUEST
  - SSH_MSG_USERAUTH_FAILURE
  - SSH_MSG_USERAUTH_SUCCESS
- authentication methods used
  - public-key, password, host-based

SSH Connection Protocol

- runs on SSH Transport Layer Protocol
- assumes secure authentication connection
- used for multiple logical channels
  - SSH communications use separate channels
  - either side can open with unique id number
  - flow controlled
  - have three stages:
    - opening a channel, data transfer, closing a channel
  - four types:
    - session, x11, forwarded-tcpip, direct-tcpip

SSH Connection Protocol Exchange



Port Forwarding

- convert insecure TCP connection into a secure SSH connection
  - SSH Transport Layer Protocol establishes a TCP connection between SSH client & server
  - client traffic redirected to local SSH, travels via tunnel, then remote SSH delivers to server
- supports two types of port forwarding
  - local forwarding – hijacks selected traffic
  - remote forwarding – client acts for server